

Retreat Presentation 2026

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Replacing Aging

- Growing organs: stem cells + scaffolds
- Some organs already successfully produced in lab: e.g., bone marrow, skin.
- Scaffolds needed for 3D structure (except for stuff like bone marrow, also growing just epidermis layer of skin does not need much scaffolding). Two main approaches for scaffolding:
 - Tube scaffolds: electrically charged scaffold sprayed on tube with opposite charge, then remove the tube.
 - Bioprinting: more complex, similar to 3D printing, deposit layer by layer.
- Overview paper ([Pina et al., 2019](#)).
- **Research groups:**
 - Wake Forest Institute for Regenerative Medicine (WFIRM) [link](#). Large group doing lots of different organs.
 - Kimber group at Manchester uni ([link](#)). Did mini-kidney.

Organoids blogpost

Part 1 [blogpost](#).

- **Spheroids:** 3D structure of single cell type. **Organoids:** more cell types, some function(s) of the organ.
- Review article ([Lancaster and Knoblich, 2014](#)).
- Organs vs. organoids: ([Jensen and Little, 2023](#))
 - Points out difference between pluripotent stem cell (PSC) and tissue stem cell-derived (TSC) organoids.
 - TSC are from biopsy lineage-restricted, so should be more precise (but depend on the patient). Also applicable only for tissue where stem cells exist and can be collected (e.g., heart has no stem cells, in brain the biopsy would be too invasive).
- Making organoids: Stem cells (TSC or PSC), Matrigel (gelatinous protein mixture), growth factors/small molecules.
- **Matrigel:** gelatinous protein mixture secreted by Engelbreth-Holm-Swarm mouse sarcoma cells (cancer cells). Quite heterogeneous (batch effect). Alternatives exist, but Matrigel still dominates (is easy to use). Matrigel is produced by injecting the cancer cells into mice, waiting for the mice to grow tumors, then killing the mice and grinding up the tumors

- [Glass et al. \(2024\)](#) tried to measure heterogeneity across sites (same setup in all of them). Some aspect stayed consistent (e.g., cell composition), but lots of DE genes across organoids.
- Organoids are quite small - up to few millimeters in diameter.
- Organoids are not vascularized, and it would be challenging to make them so ([LaMontagne et al., 2022](#)) :
 - Oxygen gets in only via diffusion, up to 0.3mm.
 - So organoids have **necrotic core**. This also influence the surrounding non-necrotic cells.
 - Drug delivery normally happens via blood stream, so this aspect is missing in drug experiments on organoids.
- Organoids are lacking an immune system.
- Organoids are from 'fetal-like' cells. Developmental stage thus differ from adult, e.g. isoform usage (e.g., oxidative enzyme CYP3A switches from CYP3A7 $\rightarrow$ CYP3A4). Applies to PSC organoids only, TSC-derived organoids don't have this issue.
- Letting organoids to 'age', e.g., for 250 days: they move from fetal to postnatal stage, shown by [Gordon et al. \(2021\)](#). The paper also used methylation clock, S. Horvath is co-author.

Part 2 [blogpost](#).

Presentation notes

- Growing anything bigger than $\sim 0.2\text{mm}$ organoids (which has necrotic core already): we need functional delivery system for O₂ (the limiting factor) + nutrients, and to remove waste (mostly H⁺, also CO₂).
- O₂ diffusion limit: about $100 - 200\mu\text{m}$ when actively consumed by cells (quite more in a media when it's not consumed on a way, e.g. 5mm deep media is normal). Concentration drops quadratically for diffusion.
 - So in tissue, no cell is farther than $\sim 100\mu\text{m}$ from nearest capillary.
- This impose limit about $100 - 200\mu\text{m}$ on thickness of cell layer. Bigger organoids are either hollow (e.g. kidney nephron), or have necrotic core (cerebral organoids).

References

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